

EFFECTIVENESS OF WEIGHTING AND BOOTSTRAP IN THE ESTIMATION OF WELFARE INDICES UNDER COMPLEX SAMPLING

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Key words: Bias; bootstrap; confidence intervals; mean squared error; poverty and inequality measures; weighting.

Summary: Most poverty and inequality analysis rely on household-based surveys, such as income and expenditure surveys. Since multistage stratified sampling is used for these surveys, and coverage errors and non-response are generally part of surveys, it is essential that the data be weighted in order to obtain improved accuracy of the estimates of the measures. The first part of this paper considers the use of weighting, especially calibration/integrated weighting, in order to improve the estimation of welfare measures such as the Headcount index, Gini coefficient and Theil's index in complex surveys. The second part considers whether the bootstrap resampling technique provides reliable estimation results for poverty and inequality measures, especially for the construction of confidence intervals. Application of the bootstrap in complex sampling designs is discussed and used to estimate the bias and MSE of estimators of the welfare indices under consideration, over the different weighting techniques. Four confidence intervals for the welfare indices that are discussed under complex sampling, are the standard (asymptotic) interval, percentile interval, bootstrap- t interval and BCa interval. The latter confidence interval has not been applied previously in complex sampling. Results are given that assess the different weighting techniques as well as the performance of the bootstrap under complex sampling. These results are all obtained from

AMS: 62D05/62G09

a simulation study carried out on the Income and Expenditure Survey (IES) 2005/2006 of South Africa, using income as study variable.

1. Introduction

In a developing country such as South Africa, poverty is a major socio-economic concern and inequality is traditionally very high. Hence, it is essential that economic planning decisions be based on reliable information of the welfare situation. One way of obtaining the information is through the estimation of appropriate welfare indices, a collective name used for both poverty and inequality measures. Although these indices do not necessarily capture the full context of the welfare situation, they give a good indication of the level thereof.

Most poverty and inequality analyses rely on household based surveys that follow a complex sampling (CS) design. Let the population be divided into H strata. Within each stratum a sample of $n_h, h = 1, \dots, H$, primary sampling units (psu) is selected and from each selected psu, a sample of secondary sampling units (ssu) is selected. Weighting constitutes three stages. In stage one, let $d_{hji}, h = 1, \dots, H, j = 1, \dots, n_h, i = 1, \dots, n_{nj}$, denote the design weight assigned to the i -th ssu of the j -th psu of the h -th stratum. The second stage sees the adjustment of the design weights to compensate for non-response. Hence, the design weights of the respondents are adjusted upwards to account for the non-respondents and to ensure that the weights sum to the correct population total. The final stage of weighting involves the use of auxiliary information to adjust the non-response adjusted weights in order to compensate for under/over-representation in the sample. This stage makes use of calibration and integrated weighting and produces the final weights, w_{hji} ,

that are used in the estimation of the welfare indices. The following is a summary of the integrated weighting technique.

Define $\mathbf{a}_k$ as the M -vector of M auxiliary variable values for person k in household hh . For the person level auxiliary variables, the entries of $\mathbf{a}_k$ are the proportion of household members to which person k belongs, for the corresponding auxiliary characteristics (e.g. proportion of males, proportion of females, in household, etc.). For a household auxiliary variable, define a new variable for each category of the household auxiliary variable under consideration. Then, for determining the value for the category to which the household belongs, the entry is the inverse of the household size, and 0 for all other categories of the household variable, Neethling and Galpin (2006). The integrated weights, w_k , are formed by minimising a given distance function between the design weight, d_k , and w_k , subject to a set of constraints which imply that the integrated weights must give perfect estimates when applied to each auxiliary survey variable.

Different distance functions can be used in practice for minimising the distance between the design weight and the final weight. In this paper, the exponential distance (also called raking ratio) will be used where the integrated weights are given by

$$w_k = d_k \exp(\mathbf{a}_k' \boldsymbol{\lambda}_c / \nu_k),$$

with $\boldsymbol{\lambda}_c = (\lambda_1, \lambda_2, \dots, \lambda_M)'$ a Lagrange multiplier vector determined by the above constraints and ν_k a known positive weight unrelated to d_k .

Under the above CS design, it is quite challenging to determine the accuracy of estimators of welfare indices and to construct confidence intervals based on such estimators. In this paper we use the bootstrap in order to achieve this.

The aim of this paper is twofold:

1. To determine the effectiveness of the different weighting techniques in estimating measures such as the Headcount index, Gini coefficient and Theil's index; and
2. To determine the performance of the bootstrap as an estimation method under CS as well as its performance in the construction and evaluation of confidence intervals.

The paper commences with a discussion of the measures of poverty and inequality in the next section. Section 3 contains a brief discussion of the bootstrap resampling technique with a more elaborate discussion of the confidence intervals, viz. the standard (asymptotic) interval, the percentile interval, the bootstrap interval and the BCa interval, in section 5. The analysis is divided into two parts. The dataset, Income and Expenditure Survey 2005/2006 of South Africa, is described in section 4 and used to investigate the first aim of the paper as well as the performance of the bootstrap as an estimation method. The second part of the analysis is discussed in section 6 where confidence intervals are constructed and evaluated over the different weighting methods. Concluding remarks are given in section 7 along with some topics for further research.

2. Measures of poverty and inequality

A large number of welfare indices are available in the literature and are mostly based on household expenditure or income. Although these measures do not necessarily capture the full context of the welfare situation, they do, however, give a good indication of the level thereof. Since households differ in size, it is common practice to use per capita income/expenditure. This is obtained by dividing the total household income/expenditure by the number of household members. In this paper all measures will be based on per capita income.

To obtain the poverty measures a threshold needs to be defined below which a person/household is considered to be poor. This threshold is called a poverty line and can be either absolute or relative. In this paper the estimation of the welfare indices will be based on an absolute poverty line, defined as representing the same purchasing power parity in different countries. The most common choice of this poverty line is either the US \$1 per day or US \$2 per day of which the US \$2 per day will be used in this paper, Hoogeveen and Özler (2005).

2.1 Measures of poverty

2.1.1 Headcount index

This is one of the most widely used poverty measures giving the proportion of the population that lies below the defined poverty line. Under CS this measure is estimated as

$$\hat{P}_H = \frac{1}{\sum_S w_k} \sum_S w_k I(y_k < z). \quad (1)$$

Note that S denotes the sample, w_k and y_k the weight and per capita income, respectively, of person k , z indicates the poverty line and I is the indicator function, World Bank (2005).

2.1.2 Poverty gap index

The poverty gap index takes account of the depth of poverty by measuring the extent to which individuals fall below the poverty line, World Bank (2005). It is estimated, under CS, by

$$\hat{P}_1 = \frac{1}{\sum_S w_k} \sum_S w_k \frac{G_k}{z},$$

where G_k represents the poverty gap calculated as

$$G_k = (z - y_k) I(y_k < z).$$

Another measure that was considered, is the squared poverty gap index which is a weighted sum of poverty gaps that takes into account inequality among the poor. Since it performed similarly to the poverty gap index it is not described further here.

2.2 Measures of inequality

2.2.1 Gini coefficient

The Gini coefficient is well known and the most widely used measure of inequality. It is based on the Lorenz curve, a cumulative frequency curve that compares the income distribution of the population with the uniform distribution that represents perfect equality. The Gini coefficient is estimated under CS as

$$\hat{G} = \frac{2}{\sum_S w_k y_k} \sum_S w_k \hat{F}(y_k) y_k - 1, \quad (3)$$

where

$$\hat{F}(y) = \frac{1}{\sum_S w_k} \sum_S w_k I(y_k \leq y),$$

is a consistent estimator of the population distribution function of the income variable, y , Binder and Kovacevic (1995).

2.2.2 Generalised entropy indices

Just as the word “entropy” is used to describe the degree of disorder in a thermodynamic system, it is used here as the collective name for indices that measure the degree of inequality in a population. In CS its general form is estimated by

$$\widehat{GE}(\alpha) = \frac{1}{\alpha(\alpha-1)} \left[\frac{1}{\sum_S w_k} \left(\frac{1}{\bar{y}_w} \right)^\alpha \sum_S w_k (y_k)^\alpha - 1 \right], \quad (4)$$

with $\bar{y}_w$ the weighted mean income and α the representative weight given to the distances between incomes at different parts of the income distribution.

Two special cases are the mean log deviation (MLD) obtained when $\alpha = 0$ and Theil's T index obtained when $\alpha = 1$, Biewen and Jenkins (2006).

Remark: Other well-known measures of inequality are Atkinson's measures and the Laeken Indicators. Some work has been done on the Laeken indicators but will be reported on at a later stage.

3. The bootstrap resampling method

Let $\hat{\theta}$ denote an estimator of θ , the parameter of interest, that can be written as $\hat{\theta} = \hat{\theta}(\mathbf{y})$, where $\mathbf{y}$ denotes a study variable of interest of length n . Application of the bootstrap method to a simple random sample commences by selecting a bootstrap sample $\mathbf{y}^*$, with replacement, from the original sample $\mathbf{y}$ and calculating a bootstrap replicate on the bootstrap sample, $\hat{\theta}^* = \hat{\theta}(\mathbf{y}^*)$. This is repeated a large number of times, say B , resulting in B bootstrap replicates, $\{\hat{\theta}_b^*\}$, which are used in the estimation of variances and biases and for obtaining confidence intervals.

In this paper a CS design is considered from which fairly complex measures with complicated variance structures need to be estimated. Application of the bootstrap method to a complex sample is quite similar to the simple random sample case. Consider the following schematic representation of a general two-stage complex sample:

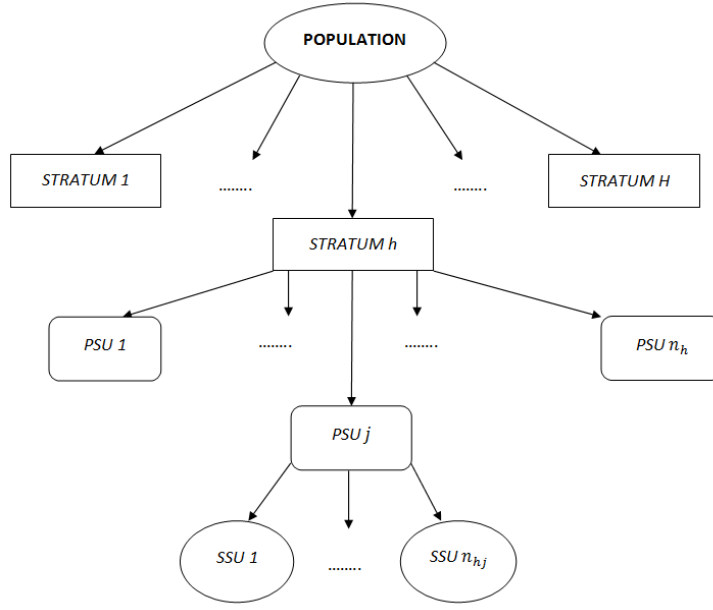


Figure 1. General two-stage CS design

Hence, the population is divided into H strata where the h -th stratum contains N_h primary sampling units (psu). A sample of n_h psu's is selected from the h -th stratum and from the j -th selected psu in the h -th stratum, a sample, of size n_{hj} , of secondary sampling units (ssu) is taken.

The bootstrap method is applied independently in each stratum where a sample of m_h psu's is taken, with replacement. Let m_{hj}^* denote the number of times the j -th psu in the h -th stratum is selected for the bootstrap sample. To compensate for some psu's appearing more than once in a bootstrap sample while others do not appear at all, define a bootstrap weight,

$$d_{hji}^* = d_{hji} \left[\left\{ 1 - \sqrt{\frac{m_h}{n_h - 1}} + \left(\sqrt{\frac{m_h}{n_h - 1}} \right) \left(\frac{n_h}{n_h - 1} \right) \cdot m_{hj}^* \right\} \right], \quad (5)$$

where d_{hji} is the design weight of the i -th ssu in the (h,j) -th psu, Rust and Rao (1996). In this paper m_h will be set equal to $n_h - 1$. As mentioned by Rust and Rao (1996), there is considerable practical benefit as well as little, if any, loss in efficiency by doing so. This choice of m_h simplifies (5) to

$$d_{hji}^* = d_{hji} \left[\left(\frac{n_h}{n_h - 1} \right) \cdot m_{hj}^* \right]. \quad (6)$$

These weights, (6), are adjusted with calibration and integrated weighting, giving $\{w_{hji}^*\}$, which are then used in the calculation of the bootstrap replicates, $\{\hat{\theta}_b^*\}$. The procedure is repeated a large number of times, B . The mean squared error (MSE) and bias of $\hat{\theta}$ is then approximated by the bootstrap MSE,

$$\widehat{MSE}_B(\hat{\theta}) = \frac{1}{B} \sum_{b=1}^B (\hat{\theta}_b^* - \hat{\theta})^2, \quad (7)$$

and the bootstrap bias,

$$\widehat{bias}_B(\hat{\theta}) = \left(\frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^* \right) - \hat{\theta}. \quad (8)$$

4. The effectiveness of weighting and bootstrap

The dataset that will be used in the analysis and that will act as surrogate population, is the Income and Expenditure survey conducted by Statistics South Africa over the period September 2005 to August 2006. This data set will be referred to as IES 2005/2006. It is carried out every five years and this IES is the most recent one available. The intention of this survey is to examine income and expenditure as well as poverty and inequality in South Africa, Statistics South Africa (2008). Note that specifically the income data is used from the IES 2005/2006.

A simulation study consisting of drawing $R = 1000$ samples from the surrogate population, was carried out and each sample followed a stratified

two-stage cluster design with the nine provinces of South Africa as strata and enumerated areas (EA), the smallest geographical areas into which the country has been divided for survey purposes, as psu's. In each selected EA, six households (HH) were selected and all household members within a selected HH were included in the sample. Differential non-response, for example the under-representation of white people living in urban areas and small households, was also simulated in the design to evaluate the weighting procedures under non-perfect circumstances which is generally found in practice. These samples are used for two purposes, to approximate the bias and MSE of estimators as well as to evaluate the performance of the bootstrap as an estimation method under CS.

The four types of weighting that will be compared, are: no weighting (None); household design weight (Design); raking ratio, integrated weighting method, based on person auxiliary variables (RR_{pp}); and raking ratio, integrated weighting method, based on person and household auxiliary variables (RR_{ph}). A discussion on integrated weighting is given in the introduction.

The auxiliary variables used in the construction of the integrated weights, are:

1. Person level auxiliary variables (pp): province (9 categories); gender (2 categories); and race (4 categories), and
2. Person and household level auxiliary variables (ph): all three person level auxiliary variables; geographic area (2 categories); and household size (8 categories).

Recall, from section 2, that the calculation of the estimates of the poverty measures requires that a poverty line, z , be defined. This line serves as threshold below which people are considered to be poor. In this study

the poverty line was set equal to an annual income of R2605 which is approximately the equivalent of \$2 per day in 2005/2006.

4.1 Effectiveness of weighting

Since the population (IES 2005/2006) is known, the quantities of interest are known. Thus, if θ denotes a particular measure of poverty or inequality, its exact value can be obtained. The measures considered in the analysis are Headcount (HC), Poverty Gap (PG), Squared Poverty Gap (PG2), Gini coefficient (Gini), Mean log deviation (MLD) and Theil's T index (Theil's T). Their exact values as found in the surrogate population are given in table 1 below.

Table 1 Population values of welfare indices

Headcount Index	Poverty Gap	Squared Poverty Gap	Gini Coefficient	Mean Log Deviation	Theil's T Index
0.29145	0.10816	0.05640	0.69456	0.94147	1.10152

For each of the $R = 1000$ samples, estimators for each of the measures are calculated using the four different methods of weighting mentioned above.

For a generic measure θ and an estimator $\hat{\theta}_r$, based on the r -th sample, true bias and MSE of that particular estimator are approximated by

$$bias(\hat{\theta}) = \left(\frac{1}{R} \sum_{r=1}^R \hat{\theta}_r \right) - \theta, \quad (9)$$

and

$$MSE(\hat{\theta}) = \frac{1}{R} \sum_{r=1}^R (\hat{\theta}_r - \theta)^2, \quad (10)$$

respectively. For each measure considered, the four estimation methods are compared in figures 2 to 5 below.

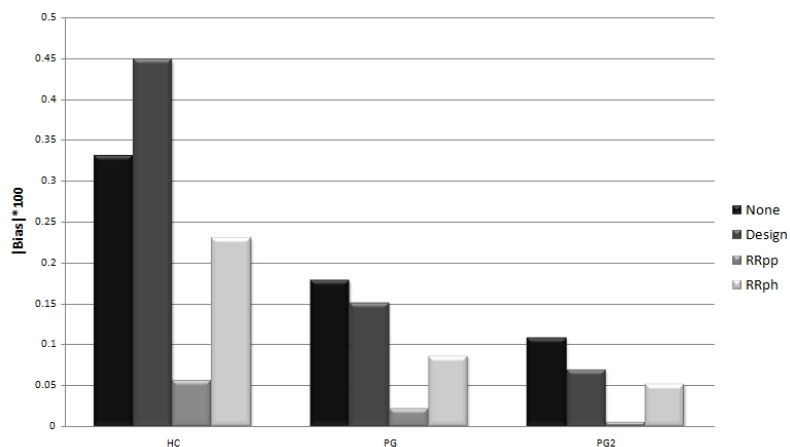


Figure 2. Bias of estimates of Poverty Measures

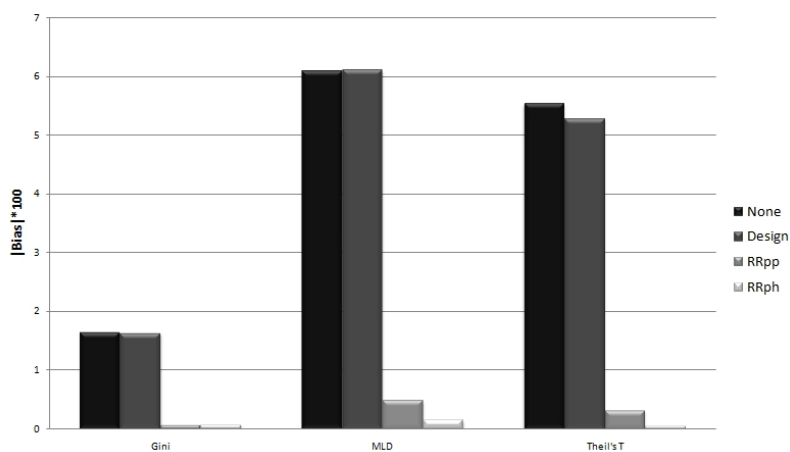


Figure 3. Bias of estimates of Inequality Measures

From figures 2 and 3 it is clear that the bias is significantly reduced by weighting the data, especially with the integrated weighting techniques, RR_{pp} and RR_{ph} .

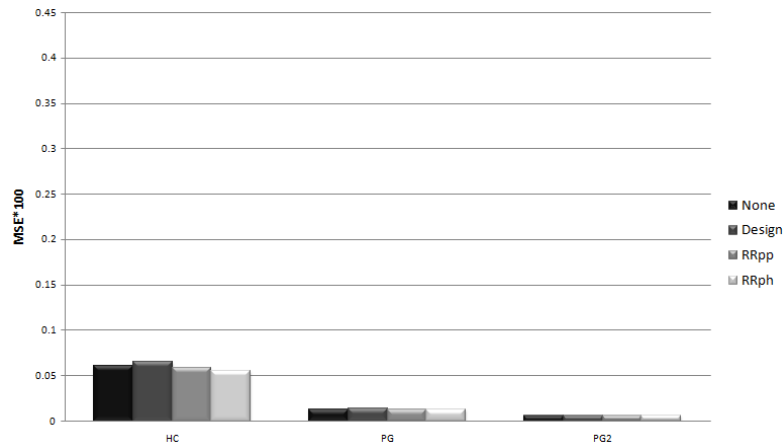


Figure 4. MSE of Estimates of Poverty Measures

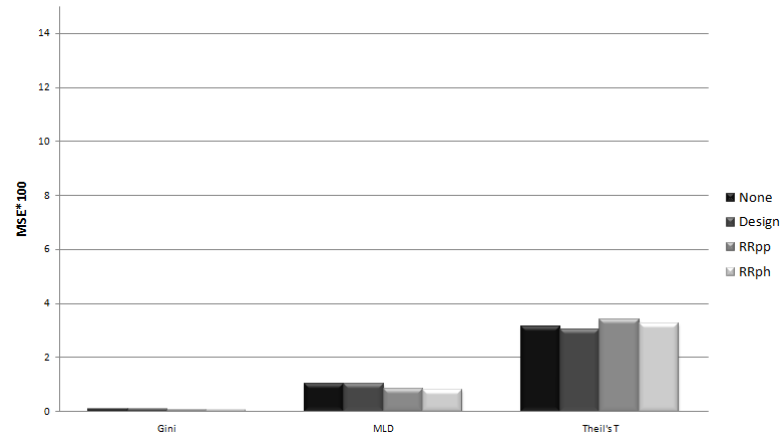


Figure 5. MSE of Estimates of Inequality Measures

In this case it is observed that the four weighting methods behave in a very similar fashion. This stable behaviour of MSE coupled with the fact that a large gain in bias reduction is obtained by integrated weighting, leads to the conclusion that integrated weighting could be highly beneficial in the estimation of welfare indices.

4.2 Effectiveness of bootstrap

In practice only a single sample is available for inferences. In order to estimate the bias and MSE of an estimator, using the bootstrap has become popular in recent years. In this study the bootstrap is applied to 200 of the $R = 1000$ simulated sample data sets to obtain an estimated bias and MSE of the estimator of the parameter of interest. This is done to evaluate the performance of the bootstrap as method of bias and MSE estimation under CS.

Consider an estimator $\hat{\theta}$ of θ based on one of the four estimation procedures (weighting procedures) and let $\hat{\theta}_r$ denote the corresponding estimator based on the r -th simulated sample data set, $r = 1, \dots, 200$. The bootstrap estimated bias and MSE, based on $B = 500$ bootstrap samples, of this estimator is given, respectively, by

$$\widehat{bias}_B(\hat{\theta}_r) = \left(\frac{1}{B} \sum_{b_r=1}^B \hat{\theta}_{b_r}^* \right) - \hat{\theta}_r,$$

and

$$\widehat{MSE}_B(\hat{\theta}_r) = \frac{1}{B} \sum_{b_r=1}^B \left(\hat{\theta}_{b_r}^* - \hat{\theta}_r \right)^2.$$

Here $\hat{\theta}_{b_r}^*$, $b = 1, \dots, B$, is the b -th estimator of $\hat{\theta}$ in the r -th sample. The performance of the bootstrap is then taken as the average of these over the 200 data sets, i.e.

$$\left| \widehat{bias}_B(\hat{\theta}) \right| = \frac{1}{200} \sum_{r=1}^{200} \left| \widehat{bias}_B(\hat{\theta}_r) \right|,$$

and

$$\widehat{MSE}_B(\hat{\theta}) = \frac{1}{200} \sum_{r=1}^{200} \widehat{MSE}_B(\hat{\theta}_r).$$

These estimators are compared to the true values, $bias(\hat{\theta})$ and $MSE(\hat{\theta})$, by defining the deviations

$$|Dev_{bias}(\hat{\theta})| = \left| \left| \widehat{bias}_B(\hat{\theta}) \right| - \left| bias(\hat{\theta}) \right| \right|, \quad (11)$$

and

$$|Dev_{MSE}(\hat{\theta})| = \left| \widehat{MSE}_B(\hat{\theta}) - MSE(\hat{\theta}) \right|. \quad (12)$$

These results are shown in figures 6 to 9 where the expressions (11) and (12) are used for two purposes:

- To evaluate the performance of the bootstrap for a particular weighting method, and
- To compare the performance of different weighting methods under application of the bootstrap.

From figures 6 and 7 it is clear that significantly better performance is obtained for the two methods using auxiliary information, RR_{pp} and RR_{ph} , compared to the two not using such information.

Although for the headcount index the highest differences between bootstrap MSE and true MSE were found for RR_{pp} and RR_{ph} , these values are still very small (< 0.00014). Thus, similar results are present in figures 8 and 9 as in figures 6 and 7.

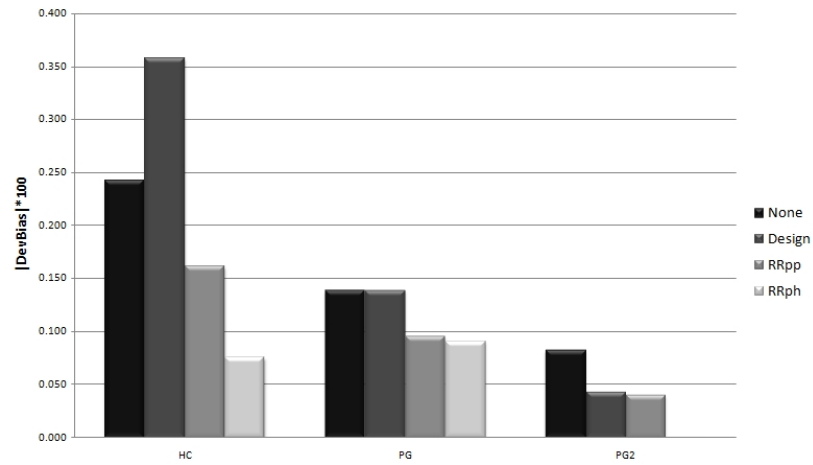


Figure 6. Difference between bootstrap estimated bias and true bias:
Poverty Measures

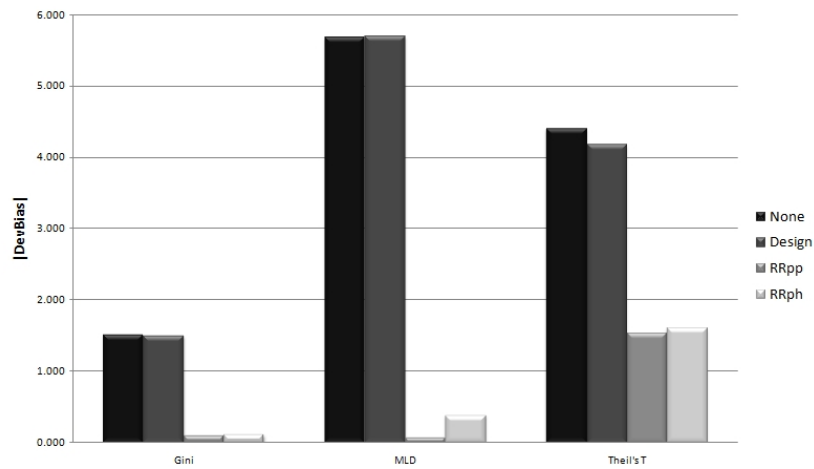


Figure 7. Difference between bootstrap estimated bias and true bias:
Inequality Measures

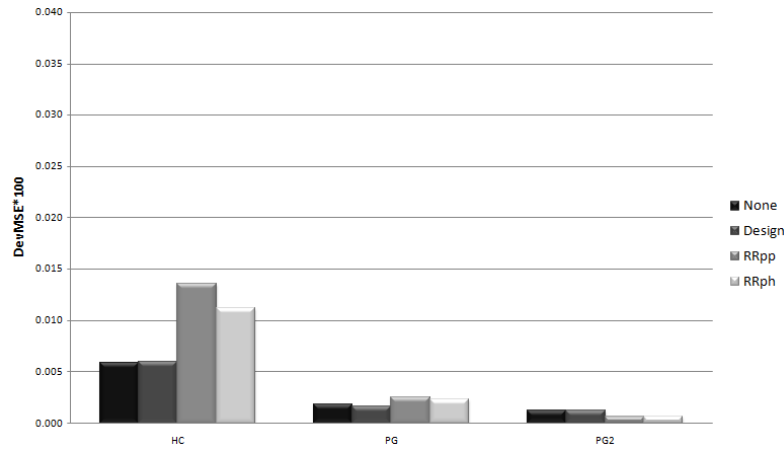


Figure 8. Difference between bootstrap estimated MSE and true MSE:
Poverty Measures

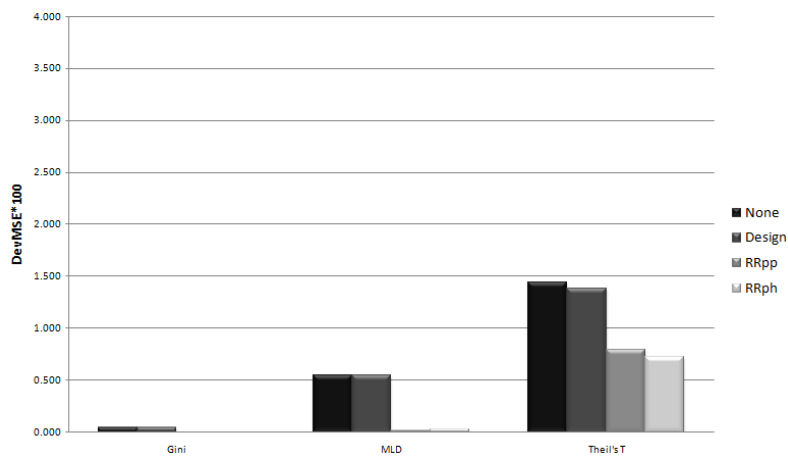


Figure 9. Difference between bootstrap estimated MSE and true MSE:
Inequality Measures

The results shown in this part of the analysis gives a clear answer to the question of whether weighting is beneficial. Most of the graphs show an improvement by using the weighted data and the best results are mostly obtained under integrated raking ratio weighting. It is also clear that the bootstrap performed well in estimating the bias and MSE of the estimators of the welfare indices.

Remark: The jackknife method was also applied in estimating the bias and MSE of the relevant estimators. Although the results are not shown here, they show a similar trend to the results given above based on the bootstrap technique.

5. Confidence intervals

In this section the construction of the confidence intervals under the CS design, as illustrated in figure 1, will be discussed. See Efron and Tibshirani (1993) for a complete discussion on the construction of bootstrap based confidence intervals under a simple random sample.

5.1 The standard (asymptotic) interval

If $\hat{\theta}$ is the estimator of the parameter of interest, θ , and assuming that $\hat{\theta}$ is approximately normally distributed with expected value θ and estimated standard error $\hat{se}$, an approximate $100(1 - \alpha)\%$ confidence interval for θ is given by

$$\left[\hat{\theta} - z^{(1-\frac{\alpha}{2})} \hat{se}(\hat{\theta}); \hat{\theta} + z^{(\frac{\alpha}{2})} \hat{se}(\hat{\theta}) \right], \quad (13)$$

where $z^{(1-\frac{\alpha}{2})}$ and $z^{(\frac{\alpha}{2})}$ are the relevant normal quantiles, Efron and Tibshirani (1993).

Under the CS design, $\hat{\theta}$ is calculated using the adjusted design weights, w_{hji} , $h = 1, \dots, H$, $j = 1, \dots, n_h$, $i = 1, \dots, n_{hj}$. The standard error,

$\widehat{se}(\widehat{\theta})$, is obtained by performing the bootstrap on the sample, as described in section 3, giving the bootstrap estimated standard error, $\widehat{se}_B(\widehat{\theta})$. Hence, the $100(1 - \alpha)\%$ confidence interval for θ will be obtained as

$$[\widehat{\theta}_{lo}; \widehat{\theta}_{up}] = \left[\widehat{\theta} - z^{(1-\frac{\alpha}{2})} \widehat{se}_B(\widehat{\theta}); \widehat{\theta} - z^{(\frac{\alpha}{2})} \widehat{se}_B(\widehat{\theta}) \right].$$

5.2 The bootstrap percentile interval

Consider the general situation where an observed sample, $\mathbf{y}$, is obtained from an unknown probability model. The bootstrap sample, $\mathbf{y}^*$, is generated from an estimated probability model and from this a bootstrap replicate is calculated, $\widehat{\theta}^* = \widehat{\theta}(\mathbf{y}^*)$. Repeating this B times, these replicates are then used to estimate the (unknown) distribution function of the estimator of the parameter of interest,

$$\widehat{G}(x) = \frac{1}{B} \sum_{b=1}^B I(\widehat{\theta}_b^* \leq x).$$

Hence, a $100(1 - \alpha)\%$ percentile interval for θ is given by Binder and Kovacevic (1995) as

$$[\widehat{\theta}_{lo}; \widehat{\theta}_{up}] = \left[\widehat{G}^{-1}\left(\frac{\alpha}{2}\right); \widehat{G}^{-1}\left(1 - \frac{\alpha}{2}\right) \right]. \quad (14)$$

Obtaining the percentile interval under a CS design through the bootstrap procedure, is quite simple. Apply the bootstrap method, independently, in each stratum as discussed in section 3. Calculate the bootstrap design weights, (6), and adjust them with calibration/integrated weighting to coincide with known population totals. These adjusted weights, w_{hji}^* , are then used in the calculation of a bootstrap replicate for each bootstrap sample, $\{\widehat{\theta}_b^*\}$, which are then sorted, $\{\widehat{\theta}_{(b)}^*\}$. $\widehat{\theta}_{lo}$ and $\widehat{\theta}_{up}$ are then taken respectively as the $B \cdot \frac{\alpha}{2}$ -th and $B \cdot \left(1 - \frac{\alpha}{2}\right)$ -th quantiles of these replicates, i.e.

$$[\widehat{\theta}_{lo}; \widehat{\theta}_{up}] = \left[\widehat{\theta}_{(B \cdot \frac{\alpha}{2})}^*; \widehat{\theta}_{(B \cdot (1 - \frac{\alpha}{2}))}^* \right]. \quad (15)$$

Note throughout it is assumed that $B \cdot \frac{\alpha}{2}$ and $B \cdot \left(1 - \frac{\alpha}{2}\right)$ are integers.

5.3 The BCa interval

The BCa confidence interval is based on the same principle as the percentile interval, but with some adjustments made to the confidence limits that correct certain shortcomings of the standard as well as percentile intervals. The interval in a simple random sample situation, as given in Efron and Tibshirani (1993), is the $100(1 - \alpha)\%$ interval given by

$$[\hat{\theta}_{lo}; \hat{\theta}_{up}] = [\hat{\theta}_{(\frac{\alpha_1}{2})}^*; \hat{\theta}_{(\frac{\alpha_2}{2})}^*],$$

where

$$\alpha_1 = \Phi \left(\hat{z}_0 + \frac{\hat{z}_0 + z(\frac{\alpha}{2})}{1 - \hat{\alpha}(\hat{z}_0 + z(\frac{\alpha}{2}))} \right), \quad (16)$$

and

$$\alpha_2 = \Phi \left(\hat{z}_0 + \frac{\hat{z}_0 + z(1 - \frac{\alpha}{2})}{1 - \hat{\alpha}(\hat{z}_0 + z(1 - \frac{\alpha}{2}))} \right). \quad (17)$$

The adjustments to the confidence limits, α_1 and α_2 , are made by the introduction of two constants: $\hat{z}_0$, bias correction, and $\hat{\alpha}$, acceleration. Since transformations, bias corrections, acceleration adjustments, etc. are frequently used to improve the performance of the standard interval, it is not surprising that the BCa interval is usually more accurate than the standard (asymptotic) interval. The difference is that the BCa confidence interval automatically incorporates these improvements, thus it is not necessary to think them through for every new application, Efron (1987).

Under the CS design, the bootstrap technique is applied independently in each of the H strata as described in section 3. This results in B bootstrap replicates of the estimator, $\{\hat{\theta}_b^*\}$, which are sorted in ascending order. The

bias correction constant, $\hat{z}_0$, is calculated as

$$\hat{z}_0 = \Phi^{-1} \left(\frac{\# \{ \hat{\theta}_b^* < \hat{\theta} \}}{B} \right),$$

with $\Phi^{-1}(\cdot)$ the inverse standard normal cumulative distribution function. To calculate the acceleration constant, the jackknife technique is applied in each of the H strata as well. Let $\hat{\theta}_{(hj)}$ be the jackknife replicate obtained after the j -th psu of the h -th stratum has been deleted. The design weights of the units are adjusted as follows:

$$d_{i(hj)} = \begin{cases} d_{hji}, & i \notin h \\ d_{hji} \cdot \frac{n_h}{(n_h - 1)}, & i \in h, i \notin j \\ 0, & i \in (h, j) \end{cases} \quad (18)$$

These jackknife weights are adjusted by calibration/integrated weighting to obtain the weights, $w_{i(hj)}$, that are used in the calculation of the jackknife replicates, $\hat{\theta}_{(hj)}$, Luus (2010). The acceleration constant, $\hat{\alpha}$, is calculated as

$$\hat{\alpha} = \frac{\sum_{h=1}^H \sum_{j=1}^{n_h} (\tilde{\theta} - \hat{\theta}_{(hj)})^3}{6 \left[\sum_{h=1}^H \sum_{j=1}^{n_h} (\tilde{\theta} - \hat{\theta}_{(hj)})^2 \right]^{\frac{3}{2}}},$$

where $\tilde{\theta} = \frac{1}{n_h} \sum_{j=1}^{n_h} \hat{\theta}_{(hj)}$. The adjusted percentiles are then obtained by substituting the values of these constants into (16) and (17), respectively. Once the adjusted confidence limits have been calculated, the BCa interval for θ is obtained by sorting the bootstrap replicates, $\{\hat{\theta}_b^*\}$, and choosing the $\frac{\alpha}{2}$ -th point of the interval as the $B \cdot \frac{\alpha_1}{2}$ -th largest value and the $\left(1 - \frac{\alpha}{2}\right)$ -th point as the $B \cdot \frac{\alpha_2}{2}$ -th largest sorted value. The $100(1 - \alpha)\%$ BCa interval is thus

given by

$$\left[\hat{\theta}_{lo}; \hat{\theta}_{up} \right] = \left[\hat{\theta}_{(B \frac{\alpha_1}{2})}^*; \hat{\theta}_{(B \frac{\alpha_2}{2})}^* \right]. \quad (19)$$

5.4 The bootstrap- t interval

Consider a “ t -statistic” $Z \equiv \frac{\hat{\theta} - \theta}{\widehat{se}}$. The bootstrap- t method uses resampling to generate B bootstrap t -statistics, $Z_1^*, \dots, Z_B^*$, where $Z_b^* = \frac{\hat{\theta}_b^* - \hat{\theta}}{\widehat{se}_b^*}$. Note that $\widehat{se}_b^*$, the bootstrap estimated standard error of $\hat{\theta}_b^*$, is calculated by resampling from the bootstrap sample, calculating the estimator for each bootstrap sample and computing the standard error of those bootstrap replicates. This results in a nested bootstrap. Once the $\{Z_b^*\}$ have been calculated, they are sorted in ascending order, $\{Z_{(b)}^*\}$, and are used to obtain the relevant percentiles for the desired confidence interval.

To obtain the bootstrap- t interval under a CS design, apply the bootstrap method as discussed in section 3 to obtain the bootstrap replicates, $\{\hat{\theta}_b^*\}$. Instead of using a nested bootstrap to obtain $\widehat{se}_b^*$, it was suggested in Rao and Wu (1987) that the jackknife method be used to save computational time. Denote the latter by $\widehat{se}_{JK}^*$. Perform the jackknife method on each bootstrap sample, $\mathbf{y}_b^*$, $b = 1, \dots, B$, independently in each stratum, by deleting the j -th psu in the h -th stratum and calculating the jackknife weights (18). These weights are then adjusted with calibration/integrated weighting to match known population totals and are then used in the calculation of the jackknife replicates, $\hat{\theta}_{(hj)}$. It follows that $\widehat{se}_{JK}^* = \sqrt{\sum_{h=1}^H \left(\frac{n_h - 1}{n_h} \right) \sum_{j=1}^{n_h} \left(\hat{\theta}_{(hj)} - \tilde{\theta}_{JK} \right)^2}$, where $\tilde{\theta}_{JK}$ is the average of the jackknife replicates. Hence, the t -statistic becomes $Z_b^* = \frac{\hat{\theta}_b^* - \hat{\theta}}{\widehat{se}_{JK}^* \left(\hat{\theta}_b^* \right)}$. These t -statistics are sorted, $\{Z_{(b)}^*\}$, and the relevant percentiles are obtained as the

$B \cdot \frac{\alpha}{2}$ -th largest and $B \cdot \left(1 - \frac{\alpha}{2}\right)$ -th largest values. Finally, the $100(1 - \alpha)\%$ bootstrap- t interval for θ is given by

$$\left[\hat{\theta}_{lo}; \hat{\theta}_{up}\right] = \left[\hat{\theta} - Z_{(B \cdot (1 - \frac{\alpha}{2}))}^* \cdot \hat{se}_B(\hat{\theta}); \hat{\theta} - Z_{(B \cdot \frac{\alpha}{2})}^* \cdot \hat{se}_B(\hat{\theta})\right], \quad (20)$$

where $\hat{se}_B(\hat{\theta})$ is the bootstrap estimated standard error of $\hat{\theta}$.

6. Confidence interval analysis

In this section the performance of the methods for confidence interval construction discussed in section 5, is discussed. In the analysis of section 4 it was found that the raking ratio integrated weighting methods, RR_{pp} and RR_{ph} , performed best. However, for completeness all four weighting methods are included in the current analysis. The IES 2005/2006 will again be used as the surrogate population with 200 of the simulated data sets used to construct the different confidence intervals.

Consider the r -th simulated data set with estimator $\hat{\theta}_r$. The standard (asymptotic) interval is given by

$$\left[\hat{\theta}_r - z_{\frac{\alpha}{2}} \hat{se}_B(\hat{\theta}_r); \hat{\theta}_r + z_{\frac{\alpha}{2}} \hat{se}_B(\hat{\theta}_r)\right],$$

the bootstrap percentile interval by

$$\left[\hat{\theta}_{(B \cdot \frac{\alpha}{2})_r}^*; \hat{\theta}_{(B \cdot (1 - \frac{\alpha}{2}))_r}^*\right],$$

the BCa interval by

$$\left[\hat{\theta}_{(B \cdot \frac{\alpha_1}{2})_r}^*; \hat{\theta}_{(B \cdot \frac{\alpha_2}{2})_r}^*\right],$$

and the bootstrap- t interval by

$$\left[\hat{\theta}_r - Z_{(B \cdot (1 - \frac{\alpha}{2}))_r}^* \cdot \hat{se}_B(\hat{\theta}_r); \hat{\theta}_r - Z_{(B \cdot \frac{\alpha_2}{2})_r}^* \cdot \hat{se}_B(\hat{\theta}_r)\right].$$

These confidence intervals will be compared according to their:

1. Non-Coverage Probability (NCP):

This measures the proportion of times that an interval does not contain the true value of the parameter of interest. It is calculated as

$$1 - \frac{1}{200} \sum_{r=1}^{200} I \left(\hat{\theta}_{r_{lo}} \leq \theta \leq \hat{\theta}_{r_{up}} \right).$$

2. Average Confidence Interval Length (ACIL):

Consider the r -th simulated data set. The length of a specific confidence interval is calculated as $l_r = (\hat{\theta}_{r_{up}} - \hat{\theta}_{r_{lo}})$. The average length of each different confidence interval is then obtained, viz. $\frac{1}{200} \sum_{r=1}^{200} l_r$.

Remark: In the figures below the darker bars represent the NCP's under the different weighting techniques for the specific confidence interval and the shaded bars represent the corresponding lengths. The dotted line represents the desired level of significance, $\alpha = 0.05$. The y -axis represents both the NCP and the ACIL's of the confidence intervals over the different weighting methods.

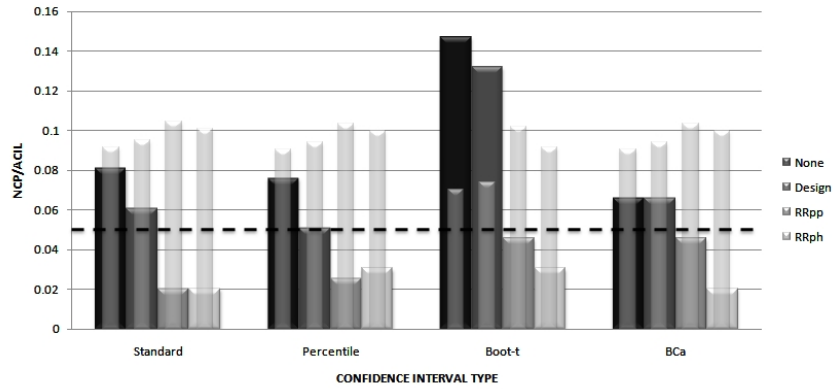


Figure 10. NCP and ACIL for the Headcount Index

Considering the NCP, figure 10 clearly shows that no weighting (None) does not do well for any of the confidence intervals. Furthermore, the standard (asymptotic) interval did not give satisfactory results for any of the estimation methods.

In the case of weighted estimators, the percentile interval for design weighting and the bootstrap- t and BCa for RR_{pp} gave excellent NCP results. Although RR_{ph} did not perform that good, it performed best under percentile and bootstrap- t intervals. The three cases gave approximately similar ACIL's.

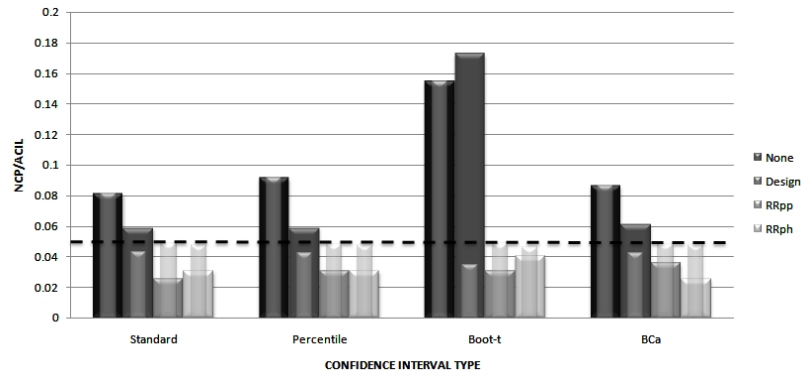


Figure 11. NCP and ACIL for the Poverty Gap Index

In figure 11, no weighting did not perform well for any of the confidence intervals. For design weighting, the standard, percentile and BCa methods gave good NCP's, all three equal ACIL's. For RR_{ph} , bootstrap- t gave a good NCP value with ACIL approximately the same as that obtained for design weighting where the latter did well. RR_{pp} did best under BCa, although not as good as RR_{ph} and design weighting where they performed well. Again in this case the ACIL was approximately comparable to that of the others.

In figure 12, no weighting can again be discarded for all confidence intervals. In this case also design weighting performed very badly over all methods. Concerning NCP, RR_{pp} and RR_{ph} both did well under all four confidence intervals with the ACIL's being approximately equal. Since the Gini coefficient is the most well-known and most often used measure of inequality, this result is very satisfactory.

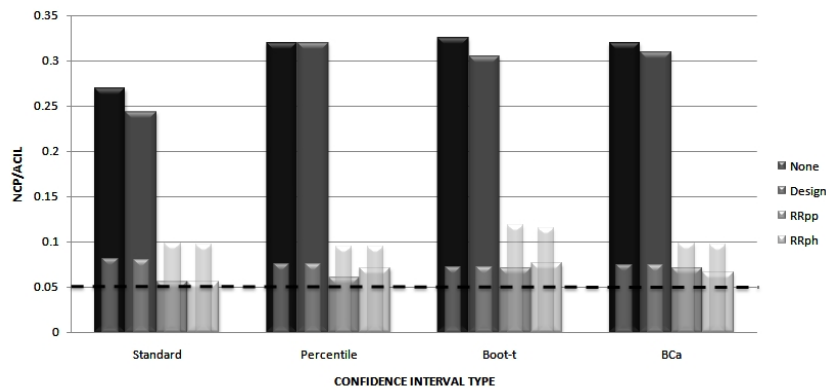


Figure 12. NCP and ACIL for the Gini Coefficient

In figure 13, both no weighting and design weighting can be discarded due to their overall poor behaviour on NCP. The BCa interval gave the most accurate NCP's for both RR_{pp} and RR_{ph} with the other three methods giving only reasonable results. For this measure of inequality the ACIL's increased substantially compared to that of the Gini measure.

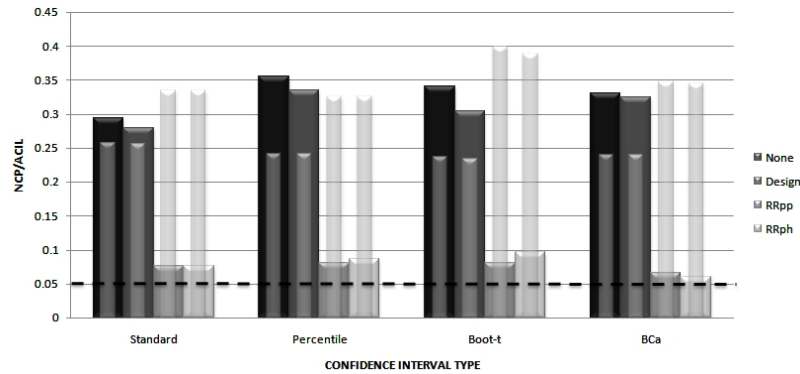


Figure 13. NCP and ACIL for the Mean Log Deviation Index

The results in this section were used to firstly assess whether weighted data must be used in the construction of confidence intervals and secondly to determine which confidence interval best suited a particular estimation method. The findings are quite conclusive, indicating that weighted data should be used, especially the integrated weighting procedures.

With fairly mixed results as to which confidence interval to use, it is seen that the bootstrap intervals performed well throughout, especially the BCa confidence interval, on the integrated, raking ratio, weighted data.

7. Conclusions and further research

The aim of this paper is twofold:

1. To determine the effectiveness of the different weighting techniques in estimating welfare measures, and
2. To determine the performance of the bootstrap as an estimation method under CS as well as its use in the construction and evaluation of confidence intervals.

The Income and Expenditure Survey 2005/2006 (IES 2005/2006) of South Africa formed the “known” population for the analysis and made it possible

to obtain the true values of welfare indices. The evaluation of four weighting techniques, viz. no weighting, design weighting and raking ratio integrated weighting on person as well as person and household auxiliary variables, was done by simulating 1000 samples from IES 2005/2006. An estimator of each welfare measure was calculated on each sample and used to estimate the bias and MSE for each measure over the different weighting techniques. It was shown that the integrated weighting techniques led to a significant reduction in bias and some improvement of the MSE.

The bootstrap is evaluated as an estimation method under CS design. This is done by applying the bootstrap to 200 of the samples generated from the IES 2005/2006 and estimating a bootstrap bias and bootstrap MSE on each sample. The average of 200 estimates is compared in terms of absolute deviation from the true bias/MSE and it is seen that the bootstrap performed well, especially under integrated weighting.

Four confidence intervals, viz. standard (asymptotic), percentile, BCa and bootstrap- t , are discussed and compared, over the four weighting procedures, in terms of NCP and ACIL's. These comparisons were used to determine the effectiveness of weighting in the construction of confidence intervals as well as deciding which confidence intervals best suit the estimation methods. It is found that weighting improved the NCP's of the different confidence intervals constructed for each of the welfare measures proving its effectiveness in this regard. The lengths generally increased slightly as the NCP's moved closer to the significance level of 0.05. The bootstrap confidence intervals perform well, especially when the integrated weighting procedures are used. Of these, the BCa confidence interval performed very well overall. This is very positive since this method has not previously been investigated under CS designs. Furthermore, it should be noted that not much research has been done

on confidence intervals for the Gini coefficient, see Qin, Rao and Wu (2010), and the good behaviour in this case of the bootstrap based intervals and in particular the BCa interval, is a very useful result.

Overall, it can be concluded that the use of weighting improved the estimated biases and MSE's of the estimators and the bootstrap is found to be effective as estimation method, also in the construction of confidence intervals for the welfare indices.

A number of further research areas can be identified. These include application of the techniques of the paper to other welfare indices, e.g. the Atkinson and quintile share ratio indices, the choice of auxiliary variables used in the integrated weighting procedures and further investigation of use of the jackknife method. These will be reported on in future research.

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